**UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS**

**GCE Ordinary Level**

# MARK SCHEME for the May/June 2010 question paper

# for the guidance of teachers

## 4024 MATHEMATICS (SYLLABUS D)

**4024/21** Paper 21, maximum raw mark 100

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UNIVERSITY of CAMBRIDGE
International Examinations

## Section A

| Qu | Answers | Mark | Comments |
|---|---|---|---|
| 1 | **(a)** f(7) = 1 as final answer | B1 | |
| | **(b)** $\frac{t-2}{5} = t$ | M1 | Forms an equation in $t$ and attempts to solve |
| | $t = -\frac{1}{2}$ | A1 | |
| | **(c)** Attempt to make $x$ the subject | M1 | |
| | $f^{-1}(x) = 5x + 2$ | A1 [5] | SC1 for $(x =)5y + 2$ |
| 2 | **(a)** $\frac{66-48}{48}$ (× 100) | M1 | |
| | 37.5% | A1 | |
| | **(b)** 130% oe soi | M1 | |
| | $\frac{19.5}{1.3}$ o.e | M1 | |
| | ($)15 | A1 | |
| | **(c)** **(i)** $88 | B1 | |
| | **(ii)** $79.20 | B1√ft | |
| | $2.8(0) cao | B1 [8] | Accept –2.8 |
| 3 | **(a)** Rectangle 13 cm by 8 cm | B1 | |
| | **(b)** **(i)** Constructs perpendicular bisector of $ZY$ | B1 | to cross rectangle |
| | Arc of circle radius 9 centre $X$ | B1 | across rectangle |
| | **(ii)** Labels the correct region | B1 | No need to shade – but must be correct |
| | **(c)** **(i)** $P$ and $Q$ correctly positioned | B1ft | |
| | **(ii)** **(a)** 42 ±1 m cao | B1 | Dep on correct $P$ and $Q$ |
| | **(b)** 107° (±2°) cao | B1 [7] | |

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| | | | |
|---|---|---|---|
| **4** | **(a)** $\frac{4(2x-1)-3(x+3)}{(x+3)(2x-1)}$ | M1 | Single fraction. Brackets not essential. Multiplies the first fraction by $(2x-1)$ and the second fraction by $(x+3)$ |
| | $\frac{8x-4-3x-9}{(x+3)(2x-1)}$ | M1 | Multiplies out the numerator with at least 1 pair of terms correct |
| | $\frac{5x-13}{(x+3)(2x-1)}$ oe as final answer | A1 | |
| | **(b)** Squares both sides of the equation | M1 | |
| | $m=\frac{k^2-3n}{2l}$ as final answer | A1 | |
| | **(c)** For num $\frac{p\pm\sqrt{q}}{r}$ | | |
| | $p = 4$ and $r = 6$ | B1 | s.o.i. or used |
| | $q = 208$ or $\sqrt{q} = 14.4...$ | B1 | |
| | $x = 3.07$,<br>$x = -1.74$ Final answers | B1<br>B1 [9] | SC1 for both 3.0 to 3.1 and –1.7 to –1.74 seen |
| **5** | **(a)** **(i)** $p = 0.5$, $q = 0.2$<br>$r = 0.3$ | B1<br>B1 | |
| | **(ii)** **(a)** 0.25 | B1 | |
| | **(b)** $0.5 \times 0.2$ seen<br>0.2 | M1<br>A1 | |
| | **(b)** **(i)** 17 | B1 | |
| | **(ii)** $78 - 54$ soi<br>$x = 8$ | M1<br>A1 [8] | Can be implied by $x + 2x + 54 = 78$ |
| **6** | **(a)** Either 136° or 44° correct<br>Other one correct | B2<br>B1ft | After B0, allow SC1 for $A\hat{C}O = 22°$, $A\hat{B}C = 68°$, $A\hat{E}C = 68°$ or for sum = 180°. |
| | **(b)** $A\hat{B}C = 68°$, $B\hat{A}C = 44°$ and $B\hat{C}A = 68°$ | B1 | |
| | Isosceles triangle | B1 [5] | Dep |

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| | | | | |
|---|---|---|---|---|
| 7 | **(a)** | Mid value used o.e.<br>Sum of (value × frequency) / 80<br>3.45 (hours) | M1<br>M1<br>A1 | |
| | **(b)** | 73, 78 | B1 | |
| | **(c)** | Correct scale, points correct and smooth curve | S1<br>P1<br>C1 | Minus 1 each error<br>P1 for 5 plots which could form ogive<br>C1 reasonable curve |
| | **(d)** | **(i)** 3.3 (hours) | B1ft | Read at 40 ft within 0.1 |
| | | **(ii)** Upper quartile and lower quartile used<br>2.5 (hours) | M1<br>A1ft[10] | Upper quartile – 2<br>ft within 0.1 |

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## Section B

| Qu | Answers | Mark | Comments |
|---|---|---|---|
| 8 | **(a)** $p = -2.6$ stated | B1 | |
| | **(b)** Scales | S1 | |
| | Five points plotted ft | P1ft | |
| | Smooth curve | C1 | Lost for ruled lines, incomplete, very thick |
| | **(c)** $x = 2.55$ to 2.65 | X1 | |
| | **(d)** **(i)** $y = x$ | L1 | |
| | **(ii)** Line drawn and attempt to read at intersect | M1 | |
| | $x = 2.4(0)$ to $2.5(0)$ | A1 | |
| | **(e)** –4 | G1 | |
| | **(f)** **(i)** Correct line drawn | T1 | Tangent of gradient part **(e)** |
| | **(ii)** (0, 12) | Y1ft | ft from *their* attempted tangent |
| | **(iii)** $y = -4x + 12$ | E1ft [12] | ft from *their* gradient and *their* intercept |
| 9 | **(a)** **(i)** $\frac{90}{360} \times \pi \times 16$ | M1 | Correct formula and 90° used |
| | +16 | M1 | Indep. Attempt to add 2 × radius |
| | 28.56 to 28.6(0) cm | A1 | |
| | **(ii)** $\frac{90}{360} \times \pi \times 8^2$ | M1 | Area of cross-section |
| | [Their $\frac{90}{360} \times \pi \times 8^2$] × $h$ = 800 soi | M1 | Indep. Forms equation |
| | $h$ = 15.9(0) to 15.92 cm | A1 | |
| | **(b)** **(i)** **(a)** $MN = 2x$ | B1 | |
| | **(b)** Area of triangle = $\frac{1}{2}$ their $(2x \times x)$ | M1 | Expect justification and a subtraction |
| | Area of sector = 16π and Subtraction | A1 | |
| | **(ii)** $20(16\pi - x^2) = 800$ | M1 | Forms equation |
| | $x^2 = 10.2...$ to 10.3 | A1 | Correct method of solution |
| | $x = 3.2(0)$ to 3.21 cm | A1 [12] | |

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| | | | | |
|---|---|---|---|---|
| **10** | **(a)** **(i)** | $140^o$ | B1 | |
| | **(ii)** | $\frac{6\times180-4\times140}{4}$ | M1 | Correct method leading to solution |
| | | or $3 \times 180 - 410$<br>or $180 - 50$ oe | | |
| | | $130^o$ | A1 | |
| | **(b)** **(i)** | $\tan 40^o = \frac{CT}{23}$ oe | M1 | |
| | | $CT = 19.29$ to $19.3(0)$ cm | A1 | |
| | **(ii)** | $73 \times 39.3$ or $50 \times 39.3$ | M1 | Accept 20 + their $CT$ for 39.3 |
| | | $\frac{1}{2}\times23\times(\text{their } CT)$ or<br>$\frac{1}{2}(20+20+\text{their } CT)\times23$ | M1 | |
| | | 2640 to 2650 $cm^2$ | A1 | |
| | **(iii)** | 10560 to 10600 | B1ft | 4 × their **(b)(ii)** |
| | **(iv)** **(a)** | 146 cm<br>79 cm | B1<br>B1ft | 40 + 2 × their **(b)(i)** rounded up |
| | | **(b)** 930 to 980 $cm^2$ cao | B1 [12] | |
| **11** | **(a)** **(i)** | $\begin{pmatrix}6\\-5\end{pmatrix}$ | B1 | Accept $\begin{matrix}6\\-5\end{matrix}$ but not 6, –5 or (6, –5) |
| | **(ii)** | Enlargement | M1 | |
| | | Scale factor $\frac{1}{2}$ | A1 | A1 and A1 not lost if transformation stated, when SC1 SC1 scored |
| | | Centre (4, 1) | A1 | |
| | **(iii)** | Shear | B1 | |
| | **(iv)** | $y = x$ (+ $c$)<br>$y = x + 1$ | M1<br>A1 | Knowing the equation has gradient 1 |
| | **(b)** **(i)** | $x$-coordinate $-q$<br>$y$-coordinate $-p$ | B1<br>B1 | SC1 for $\begin{pmatrix}-q\\-p\end{pmatrix}$ |
| | **(ii)** | $x$-coordinate $q$<br>$y$-coordinate $-p$ | B1<br>B1 | SC1 for $\begin{pmatrix}q\\-p\end{pmatrix}$ |
| | **(iii)** | $\mathbf{W} = \begin{pmatrix}-1 & 0\\0 & 1\end{pmatrix}$ | B1 [12] | |

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| | | | | |
|---|---|---|---|---|
| **12** | **(a)** **(i)** | $\mathbf{p} - \mathbf{q}$ | B1 | |
| | **(ii)** | $\frac{1}{2}(\mathbf{p} - \mathbf{q}) + \frac{1}{4}\mathbf{p}$ | M1 | Correct method |
| | | $\frac{3}{4}\mathbf{p} - \frac{1}{2}\mathbf{q}$ cao | A1 | |
| | **(b)** **(i)** **(a)** | $\frac{1}{2} \times 24 \times 17 \times \sin 55^\circ$ | M1 | |
| | | 167 to 167.5cm$^2$ | A1 | |
| | **(b)** | Attempt at cosine rule | M1 | |
| | | $XY^2 = 865 - 816 \cos 55$ | M1 | Correct formula and sign and correct algebra soi |
| | | 19.9 to 19.93 (cm) | A2 | SC1 for 396 to 397 seen |
| | **(ii)** **(a)** | $VZ^2 = 15^2 - 6^2$ | M1 | Value of 6 and correct use of Pythagoras |
| | | $VZ = 13.7$ to 13.75 cm | A1 | |
| | **(b)** | 766 cm$^3$ (Accept 762 – 766) | B1ft [12] | ft $\frac{1}{3}$ × their **(b)(i)(a)** × their **(b)(ii)(a)** |

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